Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, StandardScaler
```

Load the Dataset

```
df = pd.read_csv("/Titanic-Dataset.csv")
```

Displaying Dataset Information

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
```

Checking Missing Values

```
print(df.isnull().sum())
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

Filling Missing Numerical Values

```
for c in df.select_dtypes(include=np.number).columns:
    df[c] = df[c].fillna(df[c].median())
```

Filling Missing Categorical Values

```
for c in df.select_dtypes(exclude=np.number).columns:
    df[c] = df[c].fillna(df[c].mode()[0])
```

Label Encoding

```
le = LabelEncoder()
for c in df.select_dtypes(exclude=np.number).columns:
    df[c] = le.fit_transform(df[c])
```

Selecting Numerical Columns

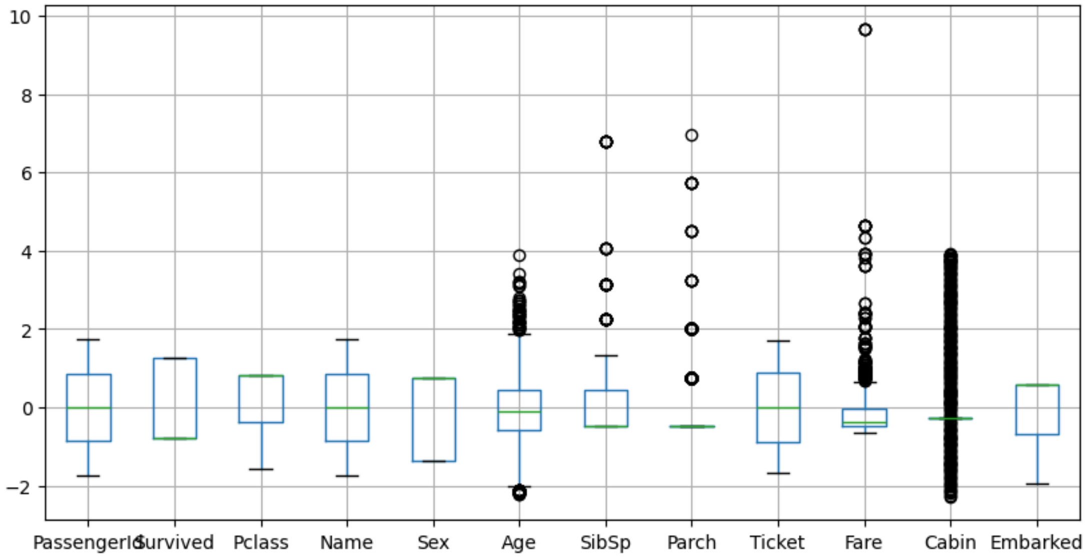
```
num_cols = df.select_dtypes(include=np.number).columns
```

Standardizing Features

```
scaler = StandardScaler()
df[num_cols] = scaler.fit_transform(df[num_cols])
```

Visualizing Outliers

```
df.boxplot(figsize=(10,5))
plt.show()
```



Calculating Quartiles

```
Q1 = df[num_cols].quantile(0.25)
Q3 = df[num_cols].quantile(0.75)
IQR = Q3 - Q1
```

Removing Outliers

```
df = df[~((df[num_cols] < (Q1 - 1.5 * IQR)) |
          (df[num_cols] > (Q3 + 1.5 * IQR))).any(axis=1)]
```

Saving the Cleaned Dataset

```
df.to_csv("cleaned_titanic.csv", index=False)
```

Completion Message

```
print("Preprocessing Completed")
```

```
Preprocessing Completed
```